

# Lab 3: Architecture Selection Justification

**System:** Self-Service Coffee Kiosk System

**Architecture Selection:** We chose Layered Architecture for the Self-Service Coffee Kiosk System.

**Reason 1:** The kiosk has a small and well-defined set of functions. Separating the touch-screen presentation, order and payment business logic, and menu data provides clear separation of concerns and keeps the design easy to understand and maintain.

**Reason 2:** The system needs to interact with specific hardware, particularly the touch screen and receipt printer, while also storing menu and pricing information. A layered structure provides a clear organization for these responsibilities and their interactions.

**Security Advantage:** Payment processing can be isolated in the business layer through the Payment Service Component. The user-interface layer does not need to directly access payment processing or stored menu data, reducing unnecessary direct access between components.

**Performance Benefit:** Keeping the kiosk's main processing within a structured local layered design avoids the distributed-service communication overhead that would be introduced by a microservices design. This is appropriate for the kiosk's limited and fixed functionality.

**Components:** Kiosk UI, Order Manager, Payment Service, Receipt Printer, and Menu Database.

**Interfaces:** UI Interface, Database Interface, Payment Service Interface, and Printer Interface.